Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/06/2024
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2024 per NSSC information (Ed., 11/6/23) NOTE: End date changed to 04/30/2023 per D. Stockton/GRC (Ed., 6/6/22) NOTE: End date changed to 06/11/2022 per NSSC information (Ed., 5/17/22) NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 (normal flames) and Round 2 (inverse flames) of Flame Design testing and are processing and analyzing the data. We are running Sphdif simulations and publishing the findings. Three journal papers were published this year and one other was submitted. One conference paper was published and presented. Kendyl Waddell defended and published his Ph.D. dissertation. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames. February 2024 Update per PI. See Bibliography for additional information. (Ed., 2/15/24) Despite a range of conditions, all flames extinguish at a critical temperature of 1130 K, a fuel-based mass flux of 0.2 g/m2-s for normal flames, and an oxygen-based mass flux of 0.68 g/m2-s for inverse flames. With this information, we have developed a simple equation to estimate the flame size (i.e., location of peak temperature) at extinction for any atmospheric-pressure ethylene spherical diffusion flame given only the reactant mass flow rate. Flame growth, which ultimately leads to radiative extinction if the critical extinction point is reached, is attributed to the natural development of the diffusion limited system as it approaches steady state and the reduction in the transport properties as the flame temperature drops due to increasing flame radiation with time (radiation-induced growth). Reference: Irace PH, Waddell KA, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst. 2023 Jan 1;39(2):1905-13. https:// Despite a narrow range of attainable conditions, flames that exhibit steady-state behavior are observed aboard the International Space Station (ISS) for up to 870 seconds, even with the constraint of a finite boundary. Steady-state flames are simulated using the numerical model for over 100,000 seconds. Reference: Irace PH, Waddell KA, Constales D, Kim M, Yablonsky G, Sunderland PB, Axelbaum RL. "On the existence of steady-state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst. 2023 Jan 1;39(2):1721-9. https://
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Frolov SM, Ivanov VS, Frolov FS, Vlasov PA, Axelbaum R, Irace PH, Yablonsky G, Waddell K. "Soot formation in spherical diffusion flames." Mathematics. 2023 Jan 4;11(2):261. https://doi.org/10.3390/math11020261 , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Waddell KA, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst. 2023 Jan 1;39(2):1905-13. https://doi.org/10.1016/j.proci.2022.07.043 [NOTE: previously reported in May 2022 as "in revision."] , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Waddell KA, Constales D, Kim M, Yablonsky G, Sunderland PB, Axelbaum RL. "On the existence of steady-state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst. 2023 Jan 1;39(2):1721-9. https://doi.org/10.1016/j.proci.2022.07.049 [NOTE: previously reported in May 2022 as "in revision."] , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Gopan A, Axelbaum RL. "An investigation of thermal radiation from laminar diffusion flames in a tri-coflow burner with central oxygen." Combustion and Flame. 2022 Aug 1;242:112158. https://doi.org/10.1016/j.combustflame.2022.112158 , Aug-2022
Articles in Peer-reviewed Journals	Krishnan S, Kumfer BM, Wu W, Li J, Nehorai A, Axelbaum RL. "An approach to thermocouple measurements that reduces uncertainties in high-temperature environments." Energy & Fuels. 2015 Mar 30;29(5):3446-55. https://doi.org/10.1021/acs.energyfuels.5b00071 , Mar-2015

Dissertations and Theses	Waddell K. "Near-limit spherical diffusion flames and cool diffusion flames." Dissertation, University of Maryland, May 2023. , May-2023
Papers from Meeting Proceedings	Waddell KA, Yablonsky G, Constales D, Axelbaum RL, Sunderland PB. "The kinetics and warm flame chemistry associated with radiative extinction of spherical diffusion flames." 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. Abstracts. 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. , Mar-2023

Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/04/2022
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2023 per D. Stockton/GRC (Ed., 6/6/22) NOTE: End date changed to 06/11/2022 per NSSC information (Ed., 5/17/22) NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 (normal flames) and Round 2 (inverse flames) of Flame Design testing and are processing the data. This includes making videos from HiBMs still images, converting video from the ops camera from ts to mp4 format, plotting the digital data for all tests, and measuring flame diameters. One journal paper was published in this year and three others were accepted. Five conference papers and posters were presented. Phillip Irace successfully defended his Ph.D. dissertation. Kendyl Waddell successfully presented his dissertation proposal. We continue to organize the results of the Flame Design ISS tests -- including flame diameter, fiber temperature, photomultiplier tubes (PMTs), radiometers, etc. Particular focus has been placed on using two-color ratio and ratio intensity pyrometry to obtain fiber temperature. We continue to plot and analyze the thermocouple data. We continue to process the HiBMs data to obtain soot volume fraction and are using the Advanced Combustion via Microgravity Experiments (ACME) images to obtain soot temperature. Fiber temperature is being used to measure flame temperature at the sooting limit. We have compiled all of our radiative extinction data, excluding extinction temperature. An analytical model of steady-state diameters of gas-fed spherical diffusion flames was improved and included in several journal paper submissions. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Other Journals or Periodicals	Irace PH, Waddell K, Constales D, Yablonsky G, Kim M, Sunderland PB, Axelbaum RL. "On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst, in revision as of May 2022. , May-2022
Articles in Other Journals or Periodicals	Irace PH, Waddell K, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst, in revision as of May 2022. , May-2022
Articles in Peer-reviewed Journals	Wang Z, Sunderland PB, Axelbaum RL. "Double blue zones in inverse and normal laminar jet diffusion flames. " Combustion and Flame. 2020 Jan; 211:253-9. https://doi.org/10.1016/j.combustflame.2019.09.014 , Jan-2020
Articles in Peer-reviewed Journals	Irace PH, Lee H, Waddell K, Tan L, Stocker DP, Sunderland PB, Axelbaum RL. "Observations of long duration microgravity spherical diffusion flames aboard the International Space Station. " Combustion and Flame. 2021 Mar 7;229:111373. https://doi.org/10.1016/j.combustflame.2021.02.019 , Mar-2021

Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/17/2020
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 of Flame Design testing and are processing the data. This includes making videos from HiBMs still images, converting video from the ops camera from ts to mp4 format, plotting the digital data for all tests, and measuring flame diameters. We have finalized the test matrix for Round 2. We are organizing the results of Flame Design Round 1 (flame diameter, fiber temperature, PMTs, radiometers, etc.). Particular focus has been placed on using two-color ratio and ratio intensity pyrometry to obtain fiber temperature. We continued to plot and analyze the thermocouple data. We are processing HiBMs data to obtain soot volume fraction and are using the Advanced Combustion via Microgravity Experiments (ACME) images to obtain soot temperature. Fiber temperature is being used to measure flame temperature at the sooting limit. We have compiled all of our radiative extinction data, excluding extinction temperature. We have performed thin-filament pyrometry for several International Space Station (ISS) flames based on color ratios and based on grayscale intensity. We have completed soot temperature measurements of ethylene and methane tri-axial flames using two-color pyrometry. We performed narrow-angle radiometer measurements of radiant emissions in ethylene tri-axial flames. Soot volume fraction measurements using color pyrometry and an absolute intensity calibration have been performed. In our research on double luminous zones, we have made our chemiluminescence images quantitative and have chosen some of them to be displayed as color contour plots of intensity at specific wavelengths. Thermocouple probing of the double luminous zones was performed. Work continued on the imaging of excited species. An analytical model of steady-state diameters of gas-fed spherical diffusion flames was created. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. Because burner heating can significantly affect the flame diameter and lifetime, we have started to implement a heat balance at the burner surface to solve for burner temperature and better conserve enthalpy. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames. We compared the simulation results with experimental results. Work is underway to update the globally implicit solver to the time-splitting method. We modified the SphDiff code to plot the radius of stoichiometric mixture fraction as a function of time, both for non-reacting and reacting flows. We added 2nd order upwind scheme. We modified the transient term in the conservation of mass equation so that it captures the density change due to temperature and concentration changes. We added an automatic temperature initialization scheme from an analytical droplet combustion model. 2. Scientific Context, Hypothesis, and Research Objectives: The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the Principal Investigator (PI), the Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Abstracts for Journals and Proceedings	Irace P, Lee HJ, Waddell K, Tan L, Stocker DP, Urban DL, Sunderland PB, Axelbaum RL. "Steady-State and Radiatively Extinguishing Spherical Diffusion Flames in Microgravity." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019

Abstracts for Journals and Proceedings	Irace P, Tan L, Waddell K, Lee HJ, Stocker DP, Urban DL, Sunderland PB, Axelbaum RL. "A Galaxy of Soot Particles." Art Competition Entry, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Wang Z, Sunderland PB, Axelbaum RL. "Dilution effects on laminar jet diffusion flame lengths." Proceedings of the Combustion Institute. 2019 Jan;37(2):1547-53. https://doi.org/10.1016/j.proci.2018.06.085 , Jan-2019
Articles in Peer-reviewed Journals	Wu W, Adeosun A, Axelbaum RL. "A new method of flame temperature measurement utilizing the acoustic emissions from laser-induced plasmas." Proceedings of the Combustion Institute. 2019 Jan;37(2):1409-15. https://doi.org/10.1016/j.proci.2018.07.096 , Jan-2019
Papers from Meeting Proceedings	Irace PH, Yang Z, Gopan A, Axelbaum RL. "Soot Formation and Radiation Heat Transfer in Tri-axial Diffusion Flames." 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. Meeting papers.11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. , Mar-2019
Papers from Meeting Proceedings	Wang Z, Sunderland PB, Axelbaum RL. "Double Luminous Zones in Laminar Jet Diffusion Flames." 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. Meeting papers.11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. 9 p. , Mar-2019

Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/12/2018
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and		

	minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Many laser-induced breakdown spectroscopy (LIBS) tests were performed. The system was optimized and the signal/data processing methodology was improved. We obtained calibration data of the relation between atomic ratio and intensity ratio of major elements (C,H,O,N) from a McKenna flat-flame burner. We reduced the burden on the transition stage by cutting down the distance from the measured point to the transition stage (for ethylene-air counter-flow burner). We measured the C/O, H/O, C/H atomic ratios along the center line of ethylene-air counter-flow diffusion flame. We used the acoustic system (LIBT) to measure the temperature profile on the center line of an ethylene-air counter-flow diffusion flame. All the results have been compared to the OPPDIF simulation and Wengdong's previous data, and we have verified that the LIBS measurement is reliable and repeatable. At Washington University in the last two months, we have performed experiments to show the PM emissions from the pure ethylene flame and comparing PM (particulate matter) data of newly bought low pressure impactor [Electrical Low-Pressure Impactor (ELPI), model ELPI+, Dekati] and Scanning Mobility Particle Spectrometry (SMPS, model 3080, TSI Inc.). A rich ethylene flame was chosen as it produces soot at a uniform rate. A co-flow diffusion flame burner was used, which comprised of two coaxial channels for the flow of fuel and air, respectively. Particle sampling is achieved with an N2-diluted sampling probe. The flow of dilution N2 through the annulus gap provides the negative pressure needed to sample the flue gas as well as cool down and dilute the particle-laden flow. The dilution ratio (DR) is controlled by N2 flow rate and the physical dimension of the annulus gap, and CO2 is chosen as the tracer gas for dilution ratio measurement. A high flow rate of dilution N2 (~45 SLPM) is applied and the dilution ratio is controlled to a range of 30 – 200. The probe samples the whole particle stream while maintaining quasi-isokinetic sampling, and is expected to cause little bias for particle measurements. The ELPI and SMPS (scanning mobility particle spectrometer) data show similar trends. The experiments were also conducted to know the efficiency of ELPI with respect to time to get the correct PSD (particle size distribution) data. It was found that after 2 minutes of measurements, ELPI have to be cleaned as it creates a virtual mode owing to the partly over filing of ELPI stages. In the future, a new dilution system will be used and PM data will be compared between SMPS and ELPI at dilution ratio of the order of 10000 using rich ethylene flames. At University of Maryland (UMD), the possible appearance of double luminous zones in normal diffusion flames was investigated in the literature and in the laboratory. Additional published flame lengths were added to our draft paper on flame lengths. Oxygen concentrations were measured in the chimney for inverse diffusion flames. A presentation on double luminous zones was made at ASGSR (American Society for Gravitational & Space Research) in Seattle. At UMD we tested IDFs (inverse diffusion flames) with CO2 as diluent in both fuel side and oxidizer side. These did not have double luminous zones. We also tested IDFs with Ar and He as diluent. These had the most remarkable double luminous zones we have seen to date. At UMD, work continues on improved imaging of species in normal and inverse diffusion flames. Interference filters have been acquired to allow the imaging of OH*, CH*, C2*, and CO2*. A Matlab script for Abel deconvolution has been modified for the resulting images. We have imaged our normal and inverse diffusion flames with various optical filters to identify the source of the blue emissions observed. We also have applied deconvolution to these images. Work has progressed on identifying the species responsible for double blue zones in normal and inverse gas jet diffusion flames. Bandpass filters allow imaging of C2* (515 nm), CO2* (455 nm), CH* (430 nm), and OH* (310 nm). The images are deconvolved, and corrections for CO2* emissions are made in the images of C2* and CH* emissions. WashU and UMD have been working together to begin welding B-type thermocouples (0.0015 in, 38 micron). These will initially be used to measure temperatures of co-flow diffusion flames at various stoichiometric mixture fraction. The injection port of the single droplet generator has been modified to allow for continuous operation with a Hencken flat flame. Droplet generation in the previous design was unstable when run with a flame. The single droplet generator will allow for the study of preferential vaporization in nonpremixed flames. We modified the design of the droplet generator to be able to maintain a steady droplet stream with flame. Drawings were sent to machine shop and parts have been modified. A new student has started at Wash U and he has been learning high-speed videography, PDPA, and manufacturing of fine-wire thermocouples. We are also developing a new method of making temperature measurements with high-speed videography. The spherical flame code has been demonstrated for the first time by the new team of Ph.D. students. Work at UMD has emphasized the numerical model. The current graduate students are now able to simulate spherical normal and inverse diffusion flames. They have implemented an improved method of choosing the initial temperature profile. The numerical modeling of spherical diffusion flames has progressed. This indicates that some normal flames of ethylene burning in air may be able to reach steady state. Furthermore, flames that continue burning with peak temperatures as low as 1000 K are observed. Previously no flames were predicted with these temperatures below 1100 K. We completed the final review of “Science Protocol on Intent to Conduct an International Collaborative Research Effort Entitled Flame Design,” to support collaborative research with Russian Scientists. Employing the principles developed by Long to support ACME (Advanced Combustion via Microgravity Experiments) research projects, WashU is developing capabilities to measure particle temperatures through direct photography.

	Based on recent findings, the new graduate student at Washington University is beginning to construct a tri-coflow burner to test a theory of applying Flame Design concepts to control heat flux. We have modified an existing burner to create a triaxial coflow flame. The inner nozzle supplies air or oxygen enriched air, the middle nozzle supplies fuel (methane) or nitrogen diluted fuel, and the outer nozzle supplies air or oxygen enriched air. The inner oxidizer creates an inner flame sheet, resulting in two distinct flame zones. The goal of this project is to maximize radiation heat transfer in methane combustion by using flame design concepts to create a soot favorable environment at the inner flame sheet and then oxidize soot particles before they are released to the ambient environment. We have been taking high speed videos of the flame sheet interactions and are preparing to take radiation measurements.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Abstracts for Journals and Proceedings	Wang Z, Sunderland PB, Axelbaum RL. "Double Blue Zones in Inverse Laminar Jet Diffusion Flames." Poster, 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. , Jul-2018
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Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/13/2017
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	October 2017 report: Beei-Huan Chao, University of Hawaii, has retired and resigned from his role as Flame Design Co-Investigator.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	The laser-induced breakdown spectroscopy (LIBS) measurement method has been improved at Washington University in St. Louis. This includes a new oscilloscope setup and preparation for better acoustic monitoring. We have synchronized the new oscilloscope with the LIBS system. We completed work on noise filtering of the acoustic measurements. We performed Matlab programming on automated acoustic aided LIBS measurements. We performed high temperature LIBS tests in a Hencken burner, LIBS tests in a McKenna burner under same equivalence ratio and different temperature, and acoustic aided LIBS measurements in a McKenna burner with varying equivalence ratio. We adapted a maximum likelihood estimation method instead of average in LIBS data processing; compared and correlated the baseline for different element emission lines; and worked on the scalar measurement. We investigated the compositional effects in laser-induced backward transfer (LIBT) measurement and investigated the screening methods of LIBT measurement by looking into the statistics. We investigated the simultaneous composition and temperature measurement using laser induced plasma. We explored the water-gas-shift equilibrium in higher hydrocarbon flames and investigated the potential of inferring molecular profiles from LIBS measured elemental compositions. We investigated the water-gas-shift equilibrium in higher hydrocarbons using detailed mechanisms for heptane and toluene. We developed a plasma model to simulate the sound wave generation as a function of laser energy input and local gas properties and to optimize the measurements using a CCD based spectrometer. Wendong Wu successfully defended his Ph.D. in May, 2017, at Washington University in St. Louis. The collaboration between Washington University and Shanghai Jiaotong University will continue in the field of laser plasma diagnostics. Graduate student Pan Du has joined the Flame Design team. At the University of Maryland, we continue to examine the effects of dilution on lengths of normal and inverse gas jet diffusion flames. The method of measuring flame lengths was refined to examine only the blue intensities in the recorded images. Both normal and inverse flames now correlate together according to the theory of Roper. We have also characterized the double flame structure visible in some inverse diffusion flames. We imaged the double flames at 310 and 430 nm. We are measuring oxygen concentrations and temperatures in and above the double flames. This will help to determine if oxygen in this region contributes to the formation of the double luminous zones. A paramagnetic oxygen analyzer is being used, which has a mole fraction range of 0-1 and an accuracy of 0.001%. At the University of Hawaii Manoa, Beei-Huan Chao has retired and resigned from his role as Flame Design Co-Investigator. His invaluable contributions to Flame Design are greatly appreciated and will be missed going forward. Prior to his retirement, progress was made on analytical studies of C/O ratio in diffusion flames. We examined extinction of spherical diffusion flames attached to a burner and revisited reactant leakage of diffusion flames. We revisited Linan's 1974 paper following the “total enthalpy” concept. The research team published two journal papers, four conference papers and abstracts, and one Ph.D. dissertation. The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 2 s drop facility.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
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Abstracts for Journals and Proceedings	Wang Z, Wu W, Sunderland PB, Axelbaum RL. "Ground-Based Research Supporting Flame Design." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
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Dissertations and Theses	El Makawy MH (Mohamed H. El Makawy). "The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames." M.S. Thesis, Ghent Univ., Department of Flow, Heat and Combustion, 54 p., Aug. 2016. , Aug-2016
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Papers from Meeting Proceedings	Wang Z, Sunderland PB, Axelbaum RL "The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames." 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 6 pp. , Apr-2017

Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2016
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan Ph.D. (University of Hawaii at Manoa) Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the Principal Investigator, the three Co-Investigators, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland (UMD), and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments • The research team published 6 conference papers and abstracts during this period. • Work at Univ. of Maryland has started a new experimental study on the effects of dilution on the stoichiometric lengths of normal and inverse gas jet diffusion flames Two new graduate students were added to this project at UMD: Zhengyang Wang (Ph.D.) and Mohamed El Makawy (M.S.). They have generated predictions of these effects from the model of Roper (1977). Stoichiometric lengths of over 250 flames have been measured. Physical constants in the model of Roper have been adjusted such that it predicts these reasonably well. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, chemical kinetics, water-gas shift reaction rates, and laser-induced breakdown spectroscopy (LIBS). The LIBS measurement method has been improved. We now use a gated spectrometer to utilize C, H, N, O atomic lines instead of ionic lines in atomic ratio measurement. We improved accuracy with acoustic aid and automated baseline correction. Shot-to-shot variation is analyzed with relative standard deviation. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. Also being investigated are ignition behavior and effects of C/O ratio in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 2 s drop facility. Scientific Context, Hypothesis, and Research Objectives The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii. Additional Accomplishments W. Wu, G. Yablonsky, R.L. Axelbaum, Observation of Water-Gas Shift Equilibrium in Diffusion Flames, Combustion and Flame, accepted. The kinetic behavior of the water-gas-shift (WGS) sub-system has been studied computationally for hydrocarbon diffusion flames in the counterflow geometry. Of specific interest was the spatial domain in which the reactions in the WGS-complex exhibit characteristics that lead to species concentrations approaching those expected from equilibrium. Two scenarios were observed for WGS equilibrium: (1) Full equilibrium in which every step of the WGS-mechanism is nearly in equilibrium; (2) “Apparent” equilibrium, which is achieved via opposing but coherent shifts away from equilibrium for the two key elementary reactions that constitute the WGS reaction. This interesting phenomenon is caused by the supply of the H radical, which participates in the forward reaction of one step and the reverse reaction of the other. This new form of apparent equilibrium differs from well-known “quasi” equilibrium, which is typically caused by the combination of a fast reversible step and slow irreversible step. Parametric domains where found in which these two scenarios are observed. For a given fuel type it was shown that the two scenarios can be separated by the location of stoichiometry. On the oxidizer side, full equilibrium of the overall WGS reaction is achieved, while on the fuel side, apparent equilibrium can occur. Z. Wang, M. El Makawy, P.B. Sunderland, B. Chao, R.L. Axelbaum, The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames, Abstract, ASGSR, Cleveland OH (2016). Many past studies have examined the stoichiometric lengths of laminar gas jet diffusion flames. These are found to be independent of pressure and burner diameter for fixed fuel mass flow rate. However, some studies report flame lengths increase when inert is added to the

fuel stream, whereas others report no change. Furthermore, measurements of effects of inert in the ambient gas are sparse, as are measurements for inverse flames. The analytical model of Roper (CNF 29:219, 1977) accounts for the full range of reactant dilution and inverse flames, but has only been tested for fuel-air systems.

This work examines the stoichiometric lengths of normal and inverse steady methane/oxygen/nitrogen jet flames burning on a 3.0 mm coflowing burner inside a glass chimney. Nitrogen dilution is such that methane mole fractions vary from 0.15 – 1 and oxygen mole fractions vary from 0.18 – 1. The measured flame lengths vary from 3 – 53 mm. Stoichiometric lengths are measured from visible emissions of the blue flame sheets. Reactant flow rates are measured with calibrated rotameters.

The predictions of the Roper model are compared with the measured flame lengths. For normal flames the agreement is best for a mass diffusivity at 25 C of 17 mm2/s and a temperature of 0.94 times the adiabatic flame temperature. This contrasts to values used by Roper for methane/air flames of 20 mm2/s and 1500 K, respectively. For the present conditions the predictions match the measurements with a R2 coefficient of determination of 0.91.

For fixed fuel flow rate, inert addition to the fuel stream is found to increase flame length, but this effect is negligible for fuel mole fractions above 0.5.

A simplified approximation of the Roper model has been obtained for normal methane/nitrogen/oxygen diffusion flames. This removes the requirement to specify adiabatic flame temperature and stoichiometric mixture fraction. The simplified model predicts flame length from the methane flow rate and mole fraction and the oxygen mole fraction.

W. Wu, R.L. Axelbaum, Interpreting Flame Structure by Mixture Fraction Measurement Using Laser-Induced Breakdown Spectroscopy, Abstract, ASGSR, Cleveland OH (2016). Measuring mixture fractions in a broad range of combustion systems with a suitable spatial and temporal resolution is critical not only for understanding the flame behavior, but also for the validation of rapidly developing numerical models. Instead of deriving the mixture fraction from measured molecular profile, a new approach of obtaining such quantities from atomic ratios measured by laser induced breakdown spectroscopy (LIBS) is introduced in this study. The atomic ratios for C, N, H, and O (e.g., C/O, H/O, and N/O) for the LIBS system were first calibrated in an ethylene-air premixed flat flame, where the equivalence ratio of the pre-mixture was varied from 0.60 to 1.80. Then the atomic ratio distributions along the centerline of an ethylene-air counter-flow diffusion flame were measured to demonstrate the capability of LIBS to measure atomic ratios between different elements throughout the entire flame. The elemental mass fractions for C, H, N, O can be thus obtained from atomic ratio profiles. The mixture fraction distribution was calculated from elemental mass fractions. Numerical results were compared to the experimental results and showed excellent agreement. The results of this work demonstrate that the LIBS technique holds promise as a simple tool for measuring atomic ratios and mixture fraction in diffusion flames.

Practical Applications. This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.

Plans for Upcoming Year Including Use of NASA Facilities. In the coming year we plan to continue with our publications of experimental and numerical work. We plan to perform additional simulations at Washington Univ., Univ. of Hawaii, and Univ. of Maryland. We plan to continue efforts to advance to flight tests and spherical burner development.

Bibliography Type:	Description: (Last Updated: 02/16/2024)
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Abstracts for Journals and Proceedings	Wu W, Axelbaum RL. "Interpreting Flame Structure by Mixture Fraction Measurement Using Laser-Induced Breakdown Spectroscopy," 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Wu W, Yablonsky G, Axelbaum RL. "Observation of water–gas shift equilibrium in diffusion flames." Combustion and Flame. 2016 Nov;173:57-64. http://dx.doi.org/10.1016/j.combustflame.2016.07.025 , Nov-2016
Papers from Meeting Proceedings	Wu W, Axelbaum RL. "Interpreting Flame Structure by Conserved Scalar Measurement Using Laser-Induced Breakdown Spectroscopy." Combustion Institute Central States Meeting, Knoxville, TN, May 15-17, 2016. Combustion Institute Central States Meeting, Knoxville, TN, May 15-17, 2016. , May-2016

Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/09/2015
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan Ph.D. (University of Hawaii at Manoa) Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and		

	minimize pollutants. This effort is a collaborative effort between the Principal Investigator, the three Co-Investigators, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments • The research team published 6 conference papers and abstracts during this period. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, chemical kinetics, and laser-induced breakdown spectroscopy (LIBS). • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 5 s drop facility. Additional Accomplishments W. Wu, P.B. Sunderland, B.-H. Chao, D.L. Urban, R.L. Axelbaum, Recent Advances in Flame Design, A Spaceflight Combustion Experiment, Abstract, ASGSR, Alexandria, VA, 2015. Flame Design is a spaceflight combustion experiment that will examine several unusual aspects of spherical diffusion flames fueled by gases. Its emphasis is on soot formation and flame extinction. This experiment, and associated ground-based work, attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints, design flames that best optimize efficiency and minimize pollutants. This presentation will introduce the overall concepts of Flame Design. It will summarize observations in the NASA Glenn 2.2 s and 5 s drop facilities, which have yielded important insights into soot formation and flame extinction behavior. Detailed numerical modeling of spherical diffusion flames have aided the understanding of the observations. Recent advances in Laser-Induced Breakdown Spectroscopy (LIBS) will be discussed. LIBS allows the C/O atom ratio to be measured in flames. This diagnostic is well suited to use in spaceflight experiments.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Abstracts for Journals and Proceedings	Wu W, Axelbaum RL. "Simultaneous Measurement of Conserved Scalars in Flames Using LIBS." SciX 2015, The Great Scientific Exchange, Providence, Rhode Island, September 27-October 2, 2015. SciX 2015, The Great Scientific Exchange, Providence, Rhode Island, September 27-October 2, 2015. , Oct-2015
Abstracts for Journals and Proceedings	Wu W, Sunderland PB, Chao B-H, Urban DL, Axelbaum RL. "Recent Advances in Flame Design, A Spaceflight Combustion Experiment." To be presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Wu W, Yablonsky G, Axelbaum RL. "New Observation of Water Gas Shift Equilibrium in Flames." 9th International Conference on Chemical Kinetics, Ghent, Belgium, June 28-July 2, 2015. 9th International Conference on Chemical Kinetics, Ghent, Belgium, June 28-July 2, 2015. , Jul-2015
Abstracts for Journals and Proceedings	Lecoustre VR, Sunderland PB, Chao B-H, Axelbaum RL. "Study of Temperature Effects on Soot Precursor Formation in Nonbuoyant, Laminar Ethylene Premixed Flames." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao B-H, Axelbaum RL. "Numerical Investigations of Temperature Effects on Soot Precursors in Ethylene Premixed Flames." 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. Paper IE09. , May-2015
Papers from Meeting Proceedings	Wu W, Axelbaum RL. "Interpreting Diffusion Flame Structure by Measuring C/O Ratio Using Laser-Induced Breakdown Spectroscopy." 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. Paper IA20. , May-2015

Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/30/2015
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan Ph.D. (University of Hawaii at Manoa) Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and		

	minimize pollutants. This effort is a collaborative effort between the Principal Investigator, the three Co-Investigators, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	New project for FY2015. NOTE (Ed., 9/30/2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A) ; see that project for previous reporting.
Bibliography Type:	Description: (Last Updated: 02/16/2024)

Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/23/2014
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	12/11/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	6
No. of PhD Candidates:	1	**No. of Master' Degrees:**	5
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/11/2014 per award of new continuation grant, per NSSC (Ed., 9/30/2015) NOTE: Project extended to 9/30/2015, per D. Stocker/GRC and NSSC, pending award of new grant (Ed., 10/6/14) NOTE (Ed., 7/23/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting. NOTE: Project extended to 9/30/2014 (NSSC information/Ed., 7/2013) NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments: • The research team published 4 peer-reviewed journal articles during this period. This work is summarized below. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, and chemical kinetics. • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 5 s drop facility. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, Modeled Quenching Limits of Spherical Hydrogen Diffusion Flames, Proceedings of the Combustion Institute 34 (2013) 887-894. Hydrogen-air diffusion flames were modeled with an emphasis on kinetic extinction. The flames were one-dimensional spherical laminar diffusion flames supported by adiabatic porous burners of various diameters. Behavior of normal (H2 flowing into quiescent air) and inverse (air flowing into quiescent H2) configurations were considered using detailed H2/O2 chemistry and transport properties with updated light component diffusivities. For the same heat release rate, inverse flames were found to be smaller and 290 K hotter than normal flames. The weakest normal flame that could be achieved before quenching has an overall heat release rate of 0.25 W, compared to 1.4 W for the weakest inverse flame. There is extensive leakage of the ambient reactant for both normal and inverse flames near extinction, which results in a premixed flame regime for diffusion flames except for the smallest burners with radii on the order of 1 micron. At high flow rates H+OH(+M)?H2O(+M) contributes over 40% of the net heat release. However at flow rates approaching quenching limits, H+O2(+M)?HO2(+M) is the elementary reaction with the largest heat release rate. The corresponding author for this paper was P.B. Sunderland. F. Xia, R.L. Axelbaum, Simplifying the complexity of diffusion flames through interpretation in C/O ratio space, Computers and Mathematics with Applications, 65 (2013) 1625-1632. Understanding the structure of diffusion flames is often complicated by the dependence of flame structure on the boundary conditions, e.g. composition, temperature and flow field (e.g., strain rate in a counterflow flame.) In this paper, the local carbon-to-oxygen atom ratio (C/O ratio) is applied as a variable to interpret the flame and soot zone structures of counterflow diffusion flames from numerical results with detailed chemical kinetics and transport. Radical pool and soot precursor zones are shown to be clearly delineated in C/O ratio space. The boundary of these two zones, as well as the flame location, are shown to be independent of both stoichiometric mixture fraction (Zst) and strain rate when interpreted in C/O space. The kinetic ratio is used to study the characteristics of key chemical reactions and to identify regions of equilibrium for these reactions. The results of this paper indicate that the C/O ratio is a valuable variable for interpreting flame structure and soot precursor chemistry for diffusion flames. The corresponding author for this paper was R.L. Axelbaum. F. Xia, G.S. Yablonsky, R.L. Axelbaum, Numerical Study of Flame Structure and Soot Inception Interpreted in Carbon-to-Oxygen Atom Ratio Space, Proceedings of the Combustion Institute, 34 (2013) 1085-1091. The utility of interpreting flame results in carbon-to-oxygen atom ratio (C/O ratio) space, as opposed to physical space or mixture fraction space, is evaluated. Flame and soot zone structures of counterflow diffusion flames have been studied for C2H4 and C3H8 and interpreted in C/O ratio space as a function of stoichiometric mixture fraction (Zst). The Burke-Schumann results expressed in C/O ratio space demonstrate how a clear and direct understanding of how structure is affected by Zst can be realized in C/O ratio space because, unlike physical or mixture fraction space, the flame location is independent of stoichiometric mixture fraction. Numerical results with detailed chemical kinetics also indicate that C/O ratio space is a fundamental variable in the sense that, for a given fuel, the location of the flame zones and critical reactions is invariant with Zst and strain rate. Two zones are clearly observed, namely the radical pool and the soot precursor zone located on the fuel side of the flame. The onset threshold of soot precursors (C6H5 and C6H6) for the high temperature side of the soot precursor zone is characterized by the depletion of radicals. The role of the hydrogen radical in flame structure and soot inception is demonstrated by studying its production and consumption channels in C/O ratio space. Finally, a modified (C/O)* is given to interpret the physical meaning of C/O ratio. The numerical results in this work indicate and explain the advantages of applying C/O ratio space in the analysis of flame structure and soot precursor chemistry. The corresponding author for this paper was R.L. Axelbaum. F. Yi, R.L. Axelbaum, Stability of Spray Combustion for Water/Alcohols Mixtures in Oxygen-Enriched Air, Proceedings of the Combustion Institute, 34 (2013) 1697-1704. The feasibility of burning fuels that are heavily diluted with water was evaluated. This concept could potentially be utilized to directly generate steam for a variety of applications, including power generation and enhanced oil recovery. Two water-soluble fuels, ethanol and 1-propanol,

	were heavily diluted with water and sprayed into a swirl-stabilized burner and the blow-off limits were obtained to identify the flame stability. The blow-off limit was experimentally determined as the lowest O2 concentration where a flame could exist under a given oxidizer flow rate. For the present experimental system, stability maps were determined as a function of fuel concentration and oxidizer flow rate for different types of nozzles and both alcohols. Contours of temperature and overall fuel mass fraction in the droplets were obtained for a flame burning 15 wt% ethanol in an oxidizer of 50% O2 and 50% N2. The results revealed that preferential vaporization of ethanol over water plays an important role in enhancing flame stability. To further understand this, a model of bi-component droplet vaporization was employed for both the diffusion limit and distillation limit modes of liquid mass transfer. The results confirmed the preferential vaporization of ethanol and demonstrated that it is necessary to account for the non-ideal solution behavior of the alcohol-water mixture to appropriately model the vaporization process. The model also demonstrated that for this experimental system, droplet vaporization was better characterized by the distillation limit mode with enhanced mass transfer by convection. The corresponding author for this paper was R.L. Axelbaum. NOTE (Ed., 7/23/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Abstracts for Journals and Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Effects of Temperature and Equivalence Ratio on Soot Precursors in Rich Premixed Flames." Poster. Eighth U.S. National Combustion Meeting, Salt Lake City, Utah, May 2013. Eighth U.S. National Combustion Meeting, Salt Lake City, Utah, May 2013. , May-2013
Articles in Peer-reviewed Journals	Lecoustre VR, Chao BH, Sunderland PB, Axelbaum RL. "Extremely weak hydrogen flames." Combustion and Flame. 2010 Nov;157(11):2209-10. http://dx.doi.org/10.1016/j.combustflame.2010.07.024 , Nov-2010
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Modeled quenching limits of spherical hydrogen diffusion flames." Proceedings of the Combustion Institute. 2013;34(1):887-94. http://dx.doi.org/10.1016/j.proci.2012.07.029 , Jan-2013
Articles in Peer-reviewed Journals	Xia F, Axelbaum RL. "Simplifying the complexity of diffusion flames through interpretation in C/O ratio space." Computers and Mathematics with Applications. 2013 May;65(10):1625-32. http://dx.doi.org/10.1016/j.camwa.2013.01.008 , May-2013
Articles in Peer-reviewed Journals	Xia F, Yablonsky GS, Axelbaum RL. "Numerical study of flame structure and soot inception interpreted in carbon-to-oxygen atom ratio space." Proceedings of the Combustion Institute. 2013;34(1):1085-91. http://dx.doi.org/10.1016/j.proci.2012.06.043 , Jan-2013
Articles in Peer-reviewed Journals	Yi F, Axelbaum RL. "Stability of spray combustion for water/alcohols mixtures in oxygen-enriched air." Proceedings of the Combustion Institute. 2013;34(1):1697-704. http://dx.doi.org/10.1016/j.proci.2012.05.088 , Jan-2013
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame. 2012 Jan;159(1):194-9. http://dx.doi.org/10.1016/j.combustflame.2011.05.022 , Jan-2012

Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/03/2012
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor has PBS and DLU performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work and computations at University of Hawaii, and RLA performing normal gravity experiments and computations at Washington University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	The research team published three peer-reviewed journal articles during this period. Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, and chemical kinetics. Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Modeled Quenching Limits of Spherical Hydrogen Diffusion Flames." Proceedings of the Combustion Institute. (in press), as of August 2012. , Aug-2012
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame 2012 Jan;159(1):194-9. http://dx.doi.org/10.1016/j.combustflame.2011.05.022 , Jan-2012
Articles in Peer-reviewed Journals	Xia F, Yablonsky GS, Axelbaum RL. "Numerical study of flame structure and soot inception interpreted in carbon-to-oxygen atom ratio space." Proceedings of the Combustion Institute. In Press, Corrected Proof, Available online 4 July 2012. http://dx.doi.org/10.1016/j.proci.2012.06.043 , Jul-2012
Articles in Peer-reviewed Journals	Yi F, Axelbaum RL. "Stability of spray combustion for water/alcohols mixtures in oxygen-enriched air." Proceedings of the Combustion Institute. In Press, Corrected Proof, Available online 18 June 2012. http://dx.doi.org/10.1016/j.proci.2012.05.088 , Jun-2012
Papers from Meeting Proceedings	Dehnavi M, Rodenhurst MK, Chao BH, Lecoustre VR, Sunderland PB, Axelbaum RL. "A Numerical Study on the Soot Free Conditions of Spherical Diffusion Flames." Spring Technical Meeting of the Central States Section of The Combustion Institute, Dayton, OH, April 23-24, 2012. In: Central States Section of the Combustion Institute Spring Technical Meeting 2012: Combustion Fundamentals and Applications. Proceedings of a meeting held 23-24 April 2012, Dayton, Ohio, USA. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1213-1220. Printed by Curran Associates, Inc. 2012. , Aug-2012
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical Investigation of Hydrogen Diffusion Flame Quenching Limits in Microgravity." Spring Technical Meeting of the Central States Section of The Combustion Institute, Dayton, OH, April 23-24, 2012. In: Central States Section of the Combustion Institute Spring Technical Meeting 2012: Combustion Fundamentals and Applications. Proceedings of a meeting held 23-24 April 2012, Dayton, Ohio, USA. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 38-46. Printed by Curran Associates, Inc. 2012. , Aug-2012

Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/02/2011
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-I's, and staff at NASA Glenn Research Center. The division of labor has PBS and DLU performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work and computations at University of Hawaii, and RLA performing normal gravity experiments and computations at Washington University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	• The research team published three peer-reviewed journal articles during this period: Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, 2010 Sep;157(9):1745-52. http://dx.doi.org/ Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame. In Press, Corrected Proof, Available online 29 June 2011. http://dx.doi.org/ Wang Q, Chao BH. "Kinetic and radiative extinctions of spherical burner-stabilized diffusion flames." Combustion and Flame 2011 Aug;158(8):1532-41. http://dx.doi.org/ • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, and chemical kinetics. • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised during the Requirements Definition Review (RDR) for this project. The research team also provided guidance on the ACME operational requirements.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame. In Press, Corrected Proof, Available online 29 June 2011. http://dx.doi.org/10.1016/j.combustflame.2011.05.022 , Jun-2011
Articles in Peer-reviewed Journals	Wang Q, Chao BH. "Kinetic and radiative extinctions of spherical burner-stabilized diffusion flames." Combustion and Flame 2011 Aug;158(8):1532-41. http://dx.doi.org/10.1016/j.combustflame.2010.12.007 , Aug-2011
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, 2010 Sep;157(9):1745-52. http://dx.doi.org/10.1016/j.combustflame.2010.04.013 , Sep-2010
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Characterization of microgravity spherical diffusion flame sooting limits." 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 6 p. , Mar-2011
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical study of laminar spherical hydrogen diffusion flames at low flow rates." 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 6 p. , Mar-2011

Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/20/2010
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor is roughly broken up with PBS and DLU performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work and computations at University of Hawaii, and RLA performing normal gravity experiments and computations at Washington University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	• The Requirements Definition Review (RDR) for this project was held in Cleveland during May 10 – 11, 2010. This review was successful and helped to advance synergies among the four ACME experiments. • The Science Requirements Document (SRD) for this project was extensively revised in preparation for the RDR. • The operational requirements, test matrix, and success criteria for this project were extensively revised in preparation for the RDR. These appear in the ACME science requirements document. • The presence of appreciable molecular oxygen at the location of maximum temperature has been observed and modeled in non-premixed oxygen-enhanced combustion processes. This work was published in one journal paper. This is significant for understanding soot formation in flames. • A paper considering kinetic and radiative extinction of spherical diffusion flames was prepared and submitted to a journal. This paper is important because an improved understanding of extinction can lead to improved combustor stability and efficiency. • Numerical simulations of sooting limits of spherical diffusion flames have been performed. This work has been published in one conference paper. • Two students completed their Ph.D. studies on this research.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, in press, July 2010. Expected publication September 2010. , Jul-2010
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: I. The presence of appreciable oxygen at the location of maximum temperature." Combustion and Flame 2009 Nov;156(11):2145–52. http://dx.doi.org/10.1016/j.combustflame.2009.07.009 , Nov-2009
Articles in Peer-reviewed Journals	Wang Q, Chao BH. "Kinetic and Radiative Extinctions of Spherical Diffusion Flames." Combustion Theory and Modeling, submitted, as of July 2010. , Jul-2010
Dissertations and Theses	Lecoustre VR. "Numerical Investigations of Gaseous Spherical Diffusion Flames." Ph.D. Dissertation, University of Maryland, Department of Mechanical Engineering, December, 2009. , Dec-2009
Dissertations and Theses	Skeen SA. "Oxygen-Enhanced Combustion: Theory and Applications." Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, August, 2009. , Aug-2009
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical Simulations of Soot Kinetics in Spherical Diffusion Flames." Eastern States Meeting of the Combustion Institute, College Park, Maryland, May 2009. Eastern States Meeting of the Combustion Institute, College Park, Maryland, May 2009, 8 pp. , May-2009
Papers from Meeting Proceedings	Lecoustre VR, Moran CW, Sunderland PB, Chao BH, Axelbaum RL. "Experimental and Numerical Investigation of Extremely Weak Hydrogen Diffusion Flames." 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009, 12 pp. , May-2009
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Urban DL, Stocker DP, Axelbaum RL. "Numerical Simulations of Spherical Diffusion Flames at their Sooting Limits." 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 9 pp. , May-2009

Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/31/2009
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor is roughly broken up with DLU and PBS performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work at University of Hawaii, and RLA performing normal gravity experiments at Washington University. The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The		

	proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed "Flame Design" as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	New project for FY2010.
Bibliography Type:	Description: (Last Updated: 02/16/2024)

Fiscal Year:	FY 2009	**Task Last Updated:**	FY 12/31/2009
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNC06AA32A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNC06AA32A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This effort is a collaborative effort between the PI, the three Co-I's, and staff at NASA Glenn Research Center. The division of labor is roughly broken up with DLU and PBS performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work at University of Hawaii, and RLA performing normal gravity experiments at Washington University. The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The		

	proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed "Flame Design" as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	The following is a summary of the research activities during the period Oct. 2006 - Sept. 2009. 1) Numerical and Experimental Observations of Spherical Diffusion Flames. Spherical diffusion flames supported on a porous sphere were studied numerically and experimentally. Experiments were performed in 2.2 s and 5.2 s microgravity facilities. Numerical results were obtained from a Chemkin-based program. The program simulates flow from a porous sphere into a quiescent environment, yields both steady state and transient results and accounts for optically thick gas-phase radiation. The low flow velocities and long residence times in these diffusion flames lead to enhanced radiative and diffusive effects. Despite similar adiabatic flame temperatures, the measured and predicted temperatures varied by as much as 700 K. The temperature reduction correlates with flame size but characteristic flow times and Lewis number also influence temperature. The numerical results show that the ambient gas Lewis number would have a strong effect on flame temperature if the flames were steady and nonradiating. For example, a 10% decrease in Lewis number would increase the steady state flame temperature by 200 K. However, for these transient, radiating flames the effect of Lewis number is small. It was also observed that when hydrocarbon fuel is supplied from the ambient the large diffusion distances associated with these flames can lead to unusual steady state compositions near the outer boundary because decomposition products can diffuse to the outer boundary. This results in a loss of chemical enthalpy from the system but the effect on flame temperature is small. Transient predictions of flame sizes are larger than those observed in microgravity experiments. Close agreement could not be obtained without either increasing the model's thermal and mass diffusion properties by 30% or reducing mass flowrate by 25%. 2) Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity. Radiative extinction of spherical diffusion flames was investigated experimentally and numerically. The experiments involved microgravity spherical diffusion flames burning ethylene and propane at 0.98 bar. Both normal (fuel flowing into oxidizer) and inverse (oxidizer flowing into fuel) flames were studied, with nitrogen supplied to either the fuel or the oxygen. Flame conditions were chosen to ensure that the flames extinguished within the 2.2 s of available test time; thus extinction occurred during unsteady flame conditions. Diagnostics included color video and thin-filament pyrometry. The computations, which simulated flow from a porous sphere into a quiescent environment, included detailed chemistry, transport, and radiation and yielded transient results. Radiative extinction was observed experimentally and simulated numerically. Extinction time, peak temperature, and radiative loss fraction were found to be independent of flow rate except at very low flow rates. Radiative heat loss was dominated by the combustion products downstream of the flame and was found to scale with flame surface area, not volume. For large transient flames the heat release rate also scaled with surface area and thus the radiative loss fraction was largely independent of flow rate. Peak temperatures at extinction onset were about 1100 K, which is significantly lower than for kinetic extinction. An important observation of this work is that while radiative heat losses can drive transient extinction, this is not only because radiative losses are increasing with time but also because the heat release rate is falling off as the flame expands away from the burner and the reactant supply to the flame decreases. 3) Soot Inception Limits in Laminar Diffusion Flames with Application to Oxy–Fuel Combustion. For diffusion flames, the combination of oxygen enrichment and fuel dilution results in an increase in the stoichiometric mixture fraction, Zst, and alters the flame structure, i.e., the relationship between the local temperature and the local gas composition. Increasing Zst has been shown to result in the reduction or even elimination of soot. In the present work, the effects of variable Zst on soot inception are investigated in normal and inverse coflow flames, using ethylene as the fuel. Use of the inverse coflow flame underscores the validity of these concepts, since the convective field in the inverse flame directs particles into the fuel-rich region. Sooting limits based on particle luminosity are measured as a function of Zst. The sooting limit is obtained by varying the amount of inert gas until soot appears above a predefined height. For each limit flame, the adiabatic flame temperature is calculated and the flame temperature at the half-height is measured. The flame temperature at the sooting limit is found to increase with Zst for both normal and inverse flames. The effects of residence time are also investigated, and the sooting limit inception temperature is found to be dependent on fuel stream velocity for both the normal and inverse configurations. A simple model applicable to oxy–fuel combustion is presented which describes how increasing Zst results in the reduction and ultimately elimination of soot. This model assumes that soot inception can only occur in a region where critical values for species, temperature, and residence time are met. The soot inception region is shown to be bounded by two isotherms: a low-temperature boundary that is a function of residence time, and a high-temperature boundary that corresponds to the location of a critical local carbon-to-oxygen ratio. The effect of increasing Zst is to move the boundaries of the soot inception zone towards each other, until the zone is infinitely thin and thus the sooting limit is reached. By comparing the model to experimental data, a critical local C/O ratio of 0.53 and a sooting limit inception temperature of 1640 K (for a characteristic residence time of 22 ms) were determined for ethylene. 4) Characteristics of Non-Premixed Oxygen-Enhanced Combustion: I. The Presence of Appreciable Oxygen at the Location of Maximum Temperature. The presence of appreciable molecular oxygen at the location of maximum temperature has been observed in non-premixed oxygen-enhanced combustion (OEC) processes, specifically in flames having a high stoichiometric mixture fraction (Zst) produced with diluted fuel and oxygen-enrichment. For conventional fuel–air flames, key features of the flame are consistent with the flame sheet approximation (FSA). In particular, the depletion of O2 at the location of maximum temperature predicted by the FSA correlates well with the near-zero O2 concentration measured at this location for conventional fuel–air flames. In contradistinction, computational analysis with detailed kinetics demonstrates that for OEC flames at high Zst: (1) there is an appreciable concentration of O2 at the location of maximum temperature and (2) the maximum temperature is not coincident with the location of global stoichiometry, O2 depletion, or maximum heat release. We investigate these phenomena computationally in three non-premixed ethylene flames at low, moderate, and high Zst, but with equivalent adiabatic flame temperatures. Results demonstrate that the location of O2 depletion occurs in the vicinity of global stoichiometry for flames of any Zst and that the presence of appreciable O2 at the location of maximum temperature for high Zst flames is caused by a shift in the location of maximum temperature relative to the location of O2 depletion. This shifting is attributed to: (1) finite-rate

	multi-step chemistry resulting in exothermic heat release that is displaced from the location of O2 depletion and (2) the relative location of the heat release region with respect to the fuel and oxidizer boundaries in mixture fraction space. A method of superposition involving a variation of the flame sheet approximation with two heat sources is shown to be sufficient in explaining this phenomenon.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." Combustion Theory and Modeling 2007 Aug;11(4):639-52. http://dx.doi.org/10.1080/13647830601161567 , Aug-2007
Articles in Peer-reviewed Journals	Santa KJ, Chao BH, Sunderland PB, Urban DL, Stocker DP, Axelbaum RL. "Radiative extinction of gaseous spherical diffusion flames in microgravity." Combustion and Flame 2007 Dec;151(4):665-75. http://dx.doi.org/10.1016/j.combustflame.2007.08.009 , Dec-2007
Articles in Peer-reviewed Journals	Kumfer BM, Skeen SA, Axelbaum RL. "Soot inception limits in laminar diffusion flames with application to oxy–fuel combustion" Combustion and Flame 2008 Aug;154(3):546–56. http://dx.doi.org/10.1016/j.combustflame.2008.03.008 , Aug-2008
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: I. The presence of appreciable oxygen at the location of maximum temperature." Combustion and Flame 2009 Nov;15(11);2145–52. http://dx.doi.org/10.1016/j.combustflame.2009.07.009 , Nov-2009
Dissertations and Theses	Lecoustre VR. "Numerical Investigations of Gaseous Spherical Diffusion Flames." Ph.D. Dissertation, Department of Mechanical Engineering, 2009. , May-2009
Dissertations and Theses	Skeen SA. "Ph.D. Dissertation." Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, 2009. , May-2009
Dissertations and Theses	Kumfer BM. "Ph.D. Dissertation." Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, 2008. , May-2008

Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/28/2007
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNC06AA32A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNC06AA32A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effort is a collaborative effort between the three Co-PI's and staff at NASA Glenn (Dr. David Urban and Mr. Dennis Stocker). The division of labor is roughly broken up with PBS performing microgravity experiments at the NASA Glen Research Center and numerical simulations at University of Maryland, BHC performing analytical work at University of Hawaii and RLA performing normal gravity experiments at Washington University. The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial		

	performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The following is a summary of the research activities during this period. 1) Work has continued on radiative extinction of spherical diffusion flames, which are being investigated experimentally and numerically. The experiments involved microgravity spherical diffusion flames burning ethylene and propane at 0.98 bar. Both normal (fuel flowing into oxidizer) and inverse (oxidizer flowing into fuel) flames are studied, with nitrogen supplied to either the fuel or the oxygen. Flame conditions were chosen to ensure that the flames extinguished within the 2.2 s of available test time; thus extinction occurred during unsteady flame conditions. Diagnostics included color video and thin-filament pyrometry. The computations, which simulated flow from a porous sphere into a quiescent environment, include detailed transport properties, detailed chemistry and an optically thick radiative heat loss model to study the transient radiative extinction of spherical diffusion flames supported by a porous spherical burner. A flow of a reactant (fuel or oxidizer) issued from a porous sphere into a quiescent environment of the other reactant was considered in the simulation. Radiative extinction was observed experimentally and simulated numerically. Extinction time, peak temperature, and radiative loss fraction (ratio of the radiative heat loss rate to the heat generation rate) were found to be independent of flow rate except at very low flow rates. Radiative heat loss was dominated by the combustion products downstream of the flame and was found to scale with flame surface area, not volume. For large transient flames the heat release rate also scaled with surface area and thus the radiative loss fraction was largely independent of flow rate. Peak temperatures at extinction onset were about 1100 K, which is significantly lower than for kinetic extinction. One observation is that while radiative heat losses can drive transient extinction, this is not because radiative losses are increasing with time (flame size) but rather because the heat release rate is falling off as the temperature drops. 2) A new numerical study of limiting conditions for soot particle inception in spherical diffusion flames has been initiated. The flames were modeled using a one-dimensional, time accurate diffusion flame code with detailed chemistry and transport and an optically thick radiation model. Seventeen normal and inverse flames were considered, covering a wide range of stoichiometric mixture fraction, adiabatic flame temperature, and residence time. These flames were previously observed to reach their sooting limits after 2 s of microgravity. Sooting-limit diffusion flames with residence times longer than 200 ms were found to have temperatures near 1190 K where C/O = 0.6, whereas flames with shorter residence times required increased temperatures. Acetylene was found to be a reasonable surrogate for soot precursor species in these flames, having peak mole fractions of about 0.01. 3) Our previous studies have shown a linear relationship between the limit flame temperature and stoichiometric mixture fraction for coflow flames at their sooting limit. The sooting limit in coflow flames is obtained by varying the amount of inert gas until soot appears above a predefined height. In this work sooting limits based on particle luminosity were measured as a function of Zst. For each limit flame, the flame temperature was measured and defined as the limit flame temperature. The limit flame temperature increases linearly with Zst, for both normal and inverse flames. The effects of residence time have also been investigated, and the soot inception temperature was found to be dependent on fuel stream velocity for both the normal and inverse configuration. Earlier experimental results and modeling indicated that a critical carbon to oxygen (C/O) ratio exists in the fuel rich region of the flame where soot inception is no longer favorable due to the presence of oxygen containing species. We extended this model to include the effects of finite rate chemistry and thus residence time on the sooting limit. This involved considering two boundaries for the soot inception zone, one dictated by the C/O ratio and the other by the finite rate chemistry of fuel pyrolysis and soot inception. By combining these two boundaries, a simple model was developed that describes how a sooting limit is attained. By utilizing measured limit temperature data with the model we found the limit inception temperature for soot formation to be 1640 K and the critical carbon to oxygen ratio to be 0.53. 4) Performed analytical analysis of the kinetic and radiative extinction limits of a steady spherical diffusion flame supported by a porous burner. A single step, overall and irreversible reaction that follows high activation energy Arrhenius kinetics was adopted to describe the reaction rate. An optically thin radiation model was used to describe the radiative heat loss. The kinetic and radiative extinction limits were observed from the analysis. Results show that the flame is easier to extinguish when the radiation intensity is higher, as expected, and there exists a maximum radiation intensity above which steady burning is not possible.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." Combustion Theory and Modelling 2007 Aug;11(4):639-52. http://dx.doi.org/10.1080/13647830601161567 , Aug-2007
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Awards	Sunderland PB, Urban DL, Stocker DP, Chao BH, Axelbaum RL. "Third Prize, Art Competition Poster for, “Spherical Ethylene Diffusion Flame in Microgravity,“ 5th U.S. Combustion Meeting, San Diego, March, 2007." Mar-2007
Papers from Meeting Proceedings	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." 45th Aerospace Sciences Meeting, Reno, Nevada, January 8–11, 2007. 45th Aerospace Sciences Meeting, Paper No. AIAA-2007-736, 2007. , Jan-2007

Papers from Meeting Proceedings	Santa KJ, Chao BH, Sunderland PB, Urban DL, Stocker DP, Axelbaum RL. "Radiative extinction of gaseous spherical diffusion flames in microgravity." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. A06, 2007. , Mar-2007
Papers from Meeting Proceedings	Wang Q, Chao BH. "Kinetic and radiative extinctions of sperical diffusion flames." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. A08, 2007. , Mar-2007
Papers from Meeting Proceedings	Lecoustre VR, Skeen SA, Chao BH, Sunderland PB, Urban DL, Stocker DP, Axelbaum RL. "A computational and analytical investigation of sooting limitsof sperical diffusion flames." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. F23, 2007. , Mar-2007

Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/13/2006
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed from 12/31/2006 to 9/30/2006 (1/2009) NOTE: grant to be reinstated with a new grant number. 8/2006, per D. Urban		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1063		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. Recognizing the unique benefits of spherical diffusion flames, employ such flames to: 1. Obtain soot-inception limits as functions of stoichiometric mixture fraction, flow direction (direction across the flame front, i.e., towards the fuel or towards the oxidizer), amount of inert (nitrogen) and pressure for ethylene and methane flames. Identify the corresponding soot-inception temperature; 2. Determine effects of stoichiometric mixture fraction and flow direction on the phenomenon of permanently blue flames and, to this end, determine the effects of stoichiometric mixture fraction and flow direction on flame temperature, size, color and soot volume fraction; 3. Obtain extinction limits as functions of stoichiometric mixture fraction, flow direction, amount of inert and pressure for ethylene and methane flames. Identify the corresponding extinction temperature; 4. Evaluate whether this flame system can be used to obtain a pseudo-flammability limit for non-premixed flames in terms of stoichiometric mixture fraction and amount of inert. If so, determine these limits for ethylene and methane flames. Soot formation and flame extinction are problems of great concern in practical combustors. Recent normal-gravity experiments using counterflow diffusion flames have shown that permanently-blue flames (which are devoid of soot regardless of strain rate) and extinction-resistant flames can be produced for many hydrocarbon fuels by nitrogen exchange (increased stoichiometric mixture fraction). Furthermore, there is a unique stoichiometric mixture fraction wherein the flame temperature at extinction can be up to 200 K lower and the strain rate up to four times greater than that for the standard hydrocarbon/air flame with the same flame temperature. The present experiments employ microgravity conditions to allow for a unique set of experiments to clarify the underlying mechanisms responsible for flame response to stoichiometric mixture fraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As the goal of this program is to develop an understanding of combustion with the aim of reducing pollutants and increasing combustion efficiency, the results are directly applicable to terrestial power generation. In particular, with the present interest in carbon sequestration via oxygen-enriched combustion the present work is of particular importance as it identifies methods of controlling a number of pollutants through oxygen-enriched combustion.
Task Progress:	Below is a summary of the progress on this project for the year ending January 1, 2006: Flame Design benefited from several analytical and computational advances in 2005. These have yielded important insight into many experimental observations by this project. This has laid the groundwork for two journal publications, one which was submitted and one which is nearing submission. Work on the general behavior of spherical diffusion flames has demonstrated that a detailed transient numerical model (with detailed chemistry, transport, and radiation) can predict flame size and flame temperature for microgravity spherical diffusion flames. A simplified model indicates that Lewis number can have a large effect on flame temperature for microgravity diffusion flames. This work has been submitted to the Combustion Symposium. Work on radiative extinction of spherical diffusion flames has demonstrated that the detailed transient model can predict extinction times for microgravity spherical diffusion flames. The work identified several scaling relationship for extinguishing diffusion flames. This work is being finalized for an upcoming journal submission. A Nikon D100 color digital still camera has been configured for use in the 2.2 second drop tower. This provides four images per drop. The images have 6 megapixels and 12 x 3 bit depth and are among the highest-fidelity images ever recorded in the drop tower. This camera has been used to record images of 15 micron SiC fibers in flames in the 2.2 second drop tower. This has yielded qualitative thin-filament pyrometry results for extinguishing spherical diffusion flames. Eighteen successful drop tests were performed in the 2.2 second drop tower.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Sunderland PB, Urban DL, Stocker DP, Chao BH, Axelbaum RL. "Sooting limits of microgravity spherical diffusion flames in oxygen-enriched air and diluted fuel." Combustion Science and Technology. 2004 Dec;176(12):2143-2164. http://dx.doi.org/10.1080/00102200490514994 , Dec-2004
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Papers from Meeting Proceedings	Santa KJ, Chao BH, Sunderland P, Taylor JL, Urban DL, Stocker DP, Axelbaum RL. "Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, Jan. 9-12, 2006 . AIAA Paper AIAA-2006-747, 9 p. January 2006. , Jan-2006

Papers from Meeting Proceedings	Sunderland PB, Taylor JL, Santa KJ, Chao B-H, Urban DL, Stocker DP, Axelbaum RL. "Extinction of Normal and Inverse Spherical Diffusion Flames in Microgravity." Presented at 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 6 p. Abstract: http://ci.confex.com/ci/2005/techprogram/P1370.HTM , Mar-2005
Papers from Meeting Proceedings	Sun Z, Axelbaum RK, Chao B-H, Santa KJ, Sunderland PB, Urban DL, Stocker DP. "Numerical and Experimental Observations of Spherical Diffusion Flames." Presented at 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 6 p. Abstract: http://ci.confex.com/ci/2005/techprogram/P1900.HTM , Mar-2005

Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/17/2006
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed from 12/31/2006 to 9/30/2006 (1/2009) note - grant renewed for 4 more years/new grant number		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii) Sunderland, Peter B (University of Maryland) Urban, David L. (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1063		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. Recognizing the unique benefits of spherical diffusion flames, employ such flames to: 1. Obtain soot-inception limits as functions of stoichiometric mixture fraction, flow direction (direction across the flame front, i.e., towards the fuel or towards the oxidizer), amount of inert (nitrogen) and pressure for ethylene and methane flames. Identify the corresponding soot-inception temperature; 2. Determine effects of stoichiometric mixture fraction and flow direction on the phenomenon of permanently blue flames and, to this end, determine the effects of stoichiometric mixture fraction and flow direction on flame temperature, size, color and soot volume fraction; 3. Obtain extinction limits as functions of stoichiometric mixture fraction, flow direction, amount of inert and pressure for ethylene and methane flames. Identify the corresponding extinction temperature; 4. Evaluate whether this flame system can be used to obtain a pseudo-flammability limit for non-premixed flames in terms of stoichiometric mixture fraction and amount of inert. If so, determine these limits for ethylene and methane flames. Soot formation and flame extinction are problems of great concern in practical combustors. Recent normal-gravity experiments using counterflow diffusion flames have shown that permanently-blue flames (which are devoid of soot regardless of strain rate) and extinction-resistant flames can be produced for many hydrocarbon fuels by nitrogen exchange (increased stoichiometric mixture fraction). Furthermore, there is a unique stoichiometric mixture fraction wherein the flame temperature at extinction can be up to 200 K lower and the strain rate up to four times greater than that for the standard hydrocarbon/air flame with the same flame temperature. The present experiments employ microgravity conditions to allow for a unique set of experiments to clarify the underlying mechanisms responsible for flame response to stoichiometric mixture fraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As the goal of this program is to develop an understanding of combustion with the aim of reducing pollutants and increasing combustion efficiency, the results are directly applicable to terrestial power generation. In particular, with the present interest in carbon sequestration via oxygen-enriched combustion the present work is of particular importance as it identifies methods of controlling a number of pollutants through oxygen-enriched combustion.
Task Progress:	Testing in the 2.2-second drop tower has yielded sooting limits of spherical microgravity diffusion flames reacting ethylene and oxygen at various levels of dilution. The results provide a quantitative comparison of the effects on sooting limits of stoichiometric mixture fraction and convection direction. Testing in the drop tower also has identified a number of extinction limits. Normal-gravity tests involving coflowing gas jet flames have considered a similar range of stoichiometric mixture fraction for both normal and inverse conditions. These results lend support to the microgravity observations. Analytical modeling has advanced the understanding of the steady-state temperature, size, and residence times of the present spherical flames. Numerical modeling, which incorporates detailed chemistry and transport, has lent insight into the effects of radiation and unsteadiness of the present spherical flames. These results have been and are being disseminated in the publications listed in the publication section.
Bibliography Type:	Description: (Last Updated: 02/16/2024)
Articles in Peer-reviewed Journals	Sunderland, P.B., Urban, D.L., Stocker, D.P., Chao, B.-H. and Axelbaum, R.L. "Sooting Limits of Microgravity Spherical Diffusion Flames in Oxygen-Enriched Air and Diluted Fuel" Combustion Science and Technology, 176 2143-2164 , Dec-2004
Articles in Peer-reviewed Journals	Liu, S., Chao, Beei-Huan and Axelbaum, R.L. "A Theoretical Study on Soot Inception in Spherical Burner-Stabilized Diffusion Flames" Combustion and Flame 140 1-23 , Jan-2005
Papers from Meeting Proceedings	Z. Sun, R.L. Axelbaum, B.-H. Chao, P.B. Sunderland, D.L. Urban, and D.P. Stocker "Effects of Radiation Heat Loss and Lewis Number on the Temperature of Spherical Microgravity Diffusion Flames" Spring Technical Meeting of the Combustion Institute Central States Section, Austin, Proceedings, 5 pp. Mar-2004
Presentation	J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, and R.L. Axelbaum "Images of Spherical Diffusion Flames" Art Exhibit Poster, 30th International Symposium on Combustion, Chicago (2004). Aug-2004
Presentation	J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, and R.L. Axelbaum "Sooting Limits and Extinction of Spherical Diffusion Flames" Work-in-Progress Poster, 30th International Symposium on Combustion, Chicago (2004). Aug-2004

Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	FLIGHT	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed from 12/31/2006 to 9/30/2006 (1/2009) note - grant renewed for 4 more years/new grant number		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1063/NCC3-1062		
Performance Goal No.:			
Performance Goal Text:			

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Articles in Peer-reviewed Journals	Liu, S., Chao, Beei-Huan and Axelbaum, R.L. "A Theoretical Study on Soot Inception in Spherical Burner-Stabilized Diffusion Flames" Combustion and Flame 140 1-23 , Jan-2005
Papers from Meeting Proceedings	Z. Sun, R.L. Axelbaum, B.-H. Chao, P.B. Sunderland, D.L. Urban, and D.P. Stocker "Effects of Radiation Heat Loss and Lewis Number on the Temperature of Spherical Microgravity Diffusion Flames" Spring Technical Meeting of the Combustion Institute Central States Section, Austin, Proceedings, 5 pp. Mar-2004
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Presentation	J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, and R.L. Axelbaum "Sooting Limits and Extinction of Spherical Diffusion Flames" Work-in-Progress Poster, 30th International Symposium on Combustion, Chicago (2004). Aug-2004